

Quantum Field Theory

Course notes

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Units: $\hbar = c = 1$. Minkowski metric $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

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1 Introduction and Classical Field Theory

The goal is to describe quantum and relativistic phenomena for systems in which creation and annihilation of particles happen (the number of particles is not conserved). **Fields** are the fundamental quantity; **particles** are excited states of fields.

Let us confine a particle in a box of length $\Delta x = L$. Then

$$\Delta p \gtrsim \frac{\hbar}{L} \implies \Delta E \gtrsim \frac{\hbar c}{L}.$$

If $\Delta E \gtrsim 2mc^2$, the vacuum starts creating/destroying (anti)particles, which happens for

$$L \lesssim \lambda_c = \frac{\hbar}{mc},$$

the **Compton wavelength**. Two approaches to quantisation: canonical quantisation and the path integral.

Remarks. Fields Φ can be scalar (real ϕ or complex φ), vector (A_μ) or spinor (ψ). We use $\hbar = c = 1$ and Minkowski space with $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

1.1 Review: the action

The action is

$$S = \int dt L = \int dt d^3x \mathcal{L}, \quad \mathcal{L} = \mathcal{L}(\Phi, \partial_\mu \Phi),$$

with L the Lagrangian and $\mathcal{L}$ the **Lagrangian density**. (Higher derivative terms end up in instabilities of the system.) $\mathcal{L}$ must yield second-order equations of motion, be Poincaré invariant (no explicit dependence on the coordinates), and respect the symmetries of the specific problem. Varying,

$$\begin{aligned} \delta S &= \int_{\mathcal{M}} d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi} \delta \Phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \partial_\mu (\delta \Phi) \right], \\ \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \partial_\mu \delta \Phi &= \underbrace{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \delta \Phi \right)}_{\text{boundary term}} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \right) \delta \Phi. \end{aligned}$$

The boundary term can be neglected as long as $\delta \Phi|_{\partial \mathcal{M}} = 0$, so

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \right) \right] \delta \Phi,$$

and $\delta S = 0$ gives the **Euler–Lagrange equations of motion**.

Free massive scalar field.

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 = \underbrace{\frac{1}{2} \dot{\phi}^2}_{\sim T} - \underbrace{\frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2}_{\sim -V},$$

whose equation of motion is the **Klein–Gordon equation**

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0, \quad \square := \partial_\mu \partial^\mu \text{ (d'Alembertian).}$$

Complex scalar field (equivalent to two real scalar fields, $\varphi = (\phi_1 + i\phi_2)/\sqrt{2}$):

$$\mathcal{L} = (\partial_\mu \varphi)(\partial^\mu \varphi)^* - m^2 \varphi \varphi^*.$$

Varying with respect to φ^* and φ :

$$\square \varphi + m^2 \varphi = 0, \quad \square \varphi^* + m^2 \varphi^* = 0.$$

Electromagnetic vector potential. For a vector field $A_\mu = (\phi, \vec{A})$, the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ contains $\vec{E}$ and $\vec{B}$. With

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - J_\mu A^\mu, \quad J^\mu = (\rho, \vec{J}),$$

the equations of motion are $\partial_\mu F^{\mu\nu} = J^\nu$.

1.2 Hamiltonian formalism

The **conjugate momentum** and the **Hamiltonian density** are

$$\pi(x) := \frac{\partial \mathcal{L}}{\partial \dot{\Phi}}, \quad \mathcal{H}(\Phi, \pi) = \pi \dot{\Phi} - \mathcal{L},$$

and Hamilton's equations of motion read

$$\dot{\Phi} = \frac{\partial \mathcal{H}}{\partial \pi}, \quad \dot{\pi} = -\frac{\partial \mathcal{H}}{\partial \Phi} + \partial_i \left(\frac{\partial \mathcal{H}}{\partial (\partial_i \Phi)} \right).$$

1.3 Symmetries and conserved “charges”

Statement (Noether's theorem). Every continuous symmetry of the Lagrangian gives rise to a conserved current, namely a vector J^μ such that $\partial_\mu J^\mu = 0$ on-shell (imposing the equations of motion).

For an infinitesimal transformation $\delta\Phi = X(\Phi)$ that is a symmetry, $\delta\mathcal{L} = \partial_\mu F^\mu$, and

$$J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi)} \delta\Phi - F^\mu, \quad Q = \int d^3x J^0 \quad \text{with} \quad \frac{dQ}{dt} = 0.$$

Types of symmetries:

- Poincaré: Lorentz transformations and translations.
- Internal: symmetries of the field (e.g. in the weak and electromagnetic interactions).

Examples: translations give the energy-momentum tensor; Lorentz transformations give the angular momentum (which differs for spinors).

Energy-momentum tensor. Under $x^\mu \rightarrow x'^\mu = x^\mu + \varepsilon^\mu$ (active transformation, $\phi'(x') = \phi(x)$),

$$\delta\phi = \phi'(x) - \phi(x) = \phi(x - \varepsilon) - \phi(x) = -\varepsilon^\nu \partial_\nu \phi, \quad \delta\mathcal{L} = -\varepsilon^\nu \partial_\nu \mathcal{L},$$

so

$$J^\mu = \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L} \right] (-\varepsilon^\nu), \quad T^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L},$$

the **energy-momentum tensor**. It can be symmetrised ($T_{\mu\nu} \propto \delta S / \delta g^{\mu\nu}$). The conserved charges are

$$E = \int d^3x T^{00} = \int d^3x (\pi \dot{\phi} - \mathcal{L}), \quad P^i = \int d^3x T^{0i} = - \int d^3x \pi \partial_i \phi.$$

Lorentz transformations. $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ with $\Lambda^\top \eta \Lambda = \eta$ (ds^2 is invariant). Infinitesimally $\Lambda^\mu{}_\nu = \delta^\mu_\nu + \omega^\mu{}_\nu$, and $\Lambda^\top \eta \Lambda = \eta$ requires $\omega^{\mu\nu} = -\omega^{\nu\mu}$. Then

$$\begin{aligned}\delta\phi &= \phi'(x) - \phi(x) = \phi(\Lambda^{-1}x) - \phi(x) = \phi(x - \omega x) - \phi(x) = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi, \\ \delta\mathcal{L} &= -\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L} = -\partial_\mu (\omega^\mu{}_\nu x^\nu \mathcal{L}),\end{aligned}$$

which gives the conserved currents

$$(\mathcal{J}^\mu)^{\rho\sigma} = x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}.$$

For $\rho = 0$, $\sigma = i$ (boosts):

$$Q^{0i} = \int d^3x (x^0 T^{0i} - x^i T^{00}), \quad \frac{dQ^{0i}}{dt} = P^i + t \underbrace{\frac{dP^i}{dt}}_0 - \frac{d}{dt} \int d^3x x^i T^{00} = 0,$$

and for the rotations $Q^{ij} = \int d^3x (x^i T^{0j} - x^j T^{0i})$.

2 Quantisation of the Free Scalar Field

2.1 Internal symmetries (continuation)

The symmetry acts on the field. For the complex scalar field $\mathcal{L} = (\partial_\mu \varphi)(\partial^\mu \varphi)^* - m^2 \varphi \varphi^*$:

$$\varphi \rightarrow e^{i\alpha} \varphi, \quad \varphi^* \rightarrow e^{-i\alpha} \varphi^* \quad \implies \quad \delta\varphi = i\alpha\varphi, \quad \delta\varphi^* = -i\alpha\varphi^*.$$

Since $\delta\mathcal{L} = 0$ ($F^\mu = 0$),

$$J^\mu = i((\partial^\mu \varphi^*)\varphi - (\partial^\mu \varphi)\varphi^*), \quad Q = \int d^3x J^0 = i \int d^3x (\dot{\varphi}^* \varphi - \dot{\varphi} \varphi^*).$$

Passive and active transformations. ϕ is a scalar iff $\phi'(x') = \phi(x)$, and

$$\delta_P \phi = \phi(x') - \phi(x) \quad (\text{passive}), \quad \delta_A \phi = \phi'(x) - \phi(x) \quad (\text{active}).$$

2.2 The free real scalar field

A free scalar field can be seen as an infinite collection of harmonic oscillators. One way to see it: a lattice model of coupled oscillators becomes the Klein–Gordon field in the continuum limit $N \rightarrow \infty$. Another way: start from the unique Poincaré-invariant action

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2 \quad \implies \quad (\square + m^2)\phi = (\partial_t^2 - \nabla^2 + m^2)\phi = 0,$$

where ∇^2 couples the field at different points. In Fourier space,

$$\phi(\vec{x}, t) = \int \frac{d^3p}{(2\pi)^3} \phi_{\vec{p}}(t) e^{i\vec{p}\cdot\vec{x}} \quad \implies \quad \partial_t^2 \phi_{\vec{p}}(t) + \underbrace{(\vec{p}^2 + m^2)}_{\omega_{\vec{p}}^2} \phi_{\vec{p}}(t) = 0 :$$

the field is an infinite collection of decoupled harmonic oscillators. Now one just needs to quantise $\phi_{\vec{p}}(t)$ (or $\phi(\vec{x}, t)$).

2.3 Quantisation of the harmonic oscillator

$$L = \frac{\dot{q}^2}{2} - \frac{\omega^2}{2} q^2 \implies \ddot{q} + \omega^2 q = 0, \quad p = \frac{\partial L}{\partial \dot{q}} = \dot{q}, \quad H = \frac{p^2}{2} + \frac{\omega^2 q^2}{2},$$

with p and q now operators satisfying $[q, p] = i\hbar \mathbb{1}$. The “ladder” (creation and annihilation) operators

$$a = \sqrt{\frac{\omega}{2\hbar}} q + \frac{i}{\sqrt{2\omega\hbar}} p, \quad a^\dagger = \sqrt{\frac{\omega}{2\hbar}} q - \frac{i}{\sqrt{2\omega\hbar}} p, \quad [a, a^\dagger] = \mathbb{1},$$

give

$$H = \frac{\hbar\omega}{2} (aa^\dagger + a^\dagger a) = \hbar\omega \left(a^\dagger a + \frac{1}{2} \mathbb{1} \right),$$

where $a^\dagger$ to the left and a to the right is **normal ordering**. From $[H, a^\dagger] = \hbar\omega a^\dagger$ and $[H, a] = -\hbar\omega a$,

$$H(a^\dagger |n\rangle) = (E_n + \hbar\omega) a^\dagger |n\rangle, \quad H(a |n\rangle) = (E_n - \hbar\omega) a |n\rangle,$$

$$|n\rangle = \frac{1}{\sqrt{n!}} (a^\dagger)^n |0\rangle, \quad \langle n|n\rangle = 1.$$

2.4 Quantisation of the free scalar field

With $\pi_{\vec{p}}(t) = \partial \mathcal{L} / \partial \dot{\phi}_{\vec{p}}(t)$, the Hamiltonian density is

$$\mathcal{H} = \frac{\pi^2}{2} + \frac{(\nabla \phi)^2}{2} + \frac{m^2 \phi^2}{2}.$$

In the Schrödinger picture (for now) we promote $\phi_{\vec{p}}$ and $\pi_{\vec{p}}$ to (time-independent) operators,

$$\phi_{\vec{p}} = \frac{1}{\sqrt{2\omega_{\vec{p}}}} (a_{\vec{p}} + a_{-\vec{p}}^\dagger), \quad \pi_{\vec{p}} = -i\sqrt{\frac{\omega_{\vec{p}}}{2}} (a_{\vec{p}} - a_{-\vec{p}}^\dagger),$$

where the combination $a_{\vec{p}} + a_{-\vec{p}}^\dagger$ ensures that $\phi(\vec{x})$ is real ($\phi_{\vec{p}}^* = \phi_{-\vec{p}}$). The commutation relations

$$[a_{\vec{p}}, a_{\vec{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}), \quad [a_{\vec{p}}, a_{\vec{q}}] = 0 = [a_{\vec{p}}^\dagger, a_{\vec{q}}^\dagger]$$

(equivalently $[\phi_{\vec{p}}, \pi_{\vec{q}}] = i(2\pi)^3 \delta^{(3)}(\vec{p} + \vec{q})$) encode the spin statistics of the field. The **mode expansion** is

$$\begin{aligned} \phi(\vec{x}) &= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (a_{\vec{p}} + a_{-\vec{p}}^\dagger) e^{i\vec{p}\cdot\vec{x}} = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}), \\ \pi(\vec{x}) &= \int \frac{d^3 p}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\vec{p}}}{2}} (a_{\vec{p}} - a_{-\vec{p}}^\dagger) e^{i\vec{p}\cdot\vec{x}}. \end{aligned}$$

The Hamiltonian

$$H = \frac{1}{2} \int d^3 x (\pi^2 + (\nabla \phi)^2 + m^2 \phi^2) = \int \frac{d^3 p}{(2\pi)^3} \omega_{\vec{p}} \left(a_{\vec{p}}^\dagger a_{\vec{p}} + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right)$$

contains an infinite constant: $\delta^{(3)}(0) \rightarrow \infty$, and moreover $\omega_{\vec{p}} \sim p$ so $\int d^3 p \omega_{\vec{p}} \rightarrow \infty$. The vacuum $|0\rangle$, with $a_{\vec{k}} |0\rangle = 0$, is the ground state, with energy

$$E_0 = \langle 0 | H | 0 \rangle = \int \frac{d^3 p}{(2\pi)^3} \frac{(2\pi)^3 \delta^{(3)}(0)}{2} \omega_{\vec{p}}.$$

With $\delta^{(3)}(0) = \int d^3x e^{-i\vec{p}\cdot\vec{x}}|_{\vec{p}=0} = V$, the vacuum energy density is

$$\varepsilon_0 = \frac{E_0}{V} = \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} = \frac{1}{4\pi^2} \int_0^\infty dp p^2 \sqrt{p^2 + m^2} \rightarrow \infty.$$

The energy density still diverges. Partial solution: an energy cut-off, since we only care about energy differences. (In general relativity one cares about the total energy, not only differences.)

We therefore use **normal ordering**:

$$:H: = \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} a_{\vec{p}}^\dagger a_{\vec{p}} \implies \langle 0|:H:|0\rangle = 0.$$

Particles. From $[H, a_{\vec{p}}] = -\omega_{\vec{p}} a_{\vec{p}}$ and $[H, a_{\vec{p}}^\dagger] = \omega_{\vec{p}} a_{\vec{p}}^\dagger$, the states

$$|\vec{p}\rangle := \sqrt{2\omega_{\vec{p}}} a_{\vec{p}}^\dagger |0\rangle, \quad \langle \vec{p}_1 | \vec{p}_2 \rangle = 2\omega_{\vec{p}} (2\pi)^3 \delta^{(3)}(\vec{p}_1 - \vec{p}_2),$$

are momentum eigenstates, and

$$H |\vec{p}\rangle = \sqrt{2\omega_{\vec{p}}} \int \frac{d^3q}{(2\pi)^3} \omega_{\vec{q}} a_{\vec{q}}^\dagger ((2\pi)^3 \delta^{(3)}(\vec{q} - \vec{p}) + a_{\vec{p}}^\dagger a_{\vec{q}}) |0\rangle = \omega_{\vec{p}} |\vec{p}\rangle :$$

$|\vec{p}\rangle$ is an eigenstate of the Hamiltonian with eigenvalue $\omega_{\vec{p}} = E_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}$. Multi-particle states $|\vec{p}_1, \dots, \vec{p}_n\rangle \propto \prod_i (a_{\vec{p}_i}^\dagger)^{n_i} |0\rangle$ belong to the **Fock space** $\mathcal{F} = \bigoplus_n \mathcal{H}_n$, a generalisation of the Hilbert space.

3 The Complex Scalar Field

$$\mathcal{L} = \partial_\mu \varphi \partial^\mu \varphi^* - m^2 \varphi \varphi^*, \quad \varphi = \frac{\phi_1 + i\phi_2}{\sqrt{2}}, \quad \pi = \dot{\varphi}^*, \quad \pi^* = \dot{\varphi}.$$

Promoting φ and π to operators,

$$\begin{aligned} \varphi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (b_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + c_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}), \\ \pi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} i\sqrt{\frac{\omega_{\vec{p}}}{2}} (b_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}} - c_{\vec{p}} e^{i\vec{p}\cdot\vec{x}}), \end{aligned}$$

with $b_{\vec{p}} = \frac{1}{\sqrt{2}}(a_{1,\vec{p}} + ia_{2,\vec{p}})$ and $c_{\vec{p}} = \frac{1}{\sqrt{2}}(a_{1,\vec{p}} - ia_{2,\vec{p}})$.

Commutation relations:

$$\begin{aligned} [\varphi(\vec{x}), \pi(\vec{y})] &= i\delta^{(3)}(\vec{x} - \vec{y}), & [\varphi^\dagger(\vec{x}), \pi^\dagger(\vec{y})] &= i\delta^{(3)}(\vec{x} - \vec{y}), \\ [b_{\vec{p}}, b_{\vec{q}}^\dagger] &= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}), & [c_{\vec{p}}, c_{\vec{q}}^\dagger] &= (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}). \end{aligned}$$

The conserved charge due to the symmetry $\varphi \rightarrow e^{i\alpha} \varphi$ is

$$Q = i \int d^3x (\dot{\varphi}^* \varphi - \varphi^* \dot{\varphi}) = i \int d^3x (\pi \varphi - \varphi^* \pi^*),$$

which after normal ordering becomes

$$:Q: = \int \frac{d^3p}{(2\pi)^3} (c_{\vec{p}}^\dagger c_{\vec{p}} - b_{\vec{p}}^\dagger b_{\vec{p}}) = N_c - N_b,$$

and likewise (following D. Tong's QFT lecture notes)

$$:H: = \int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} (c_{\vec{p}}^\dagger c_{\vec{p}} + b_{\vec{p}}^\dagger b_{\vec{p}}), \quad P^i = \int \frac{d^3p}{(2\pi)^3} p^i (b_{\vec{p}}^\dagger b_{\vec{p}} + c_{\vec{p}}^\dagger c_{\vec{p}}).$$

3.1 Heisenberg picture: causality and propagators

From the Schrödinger to the Heisenberg picture,

$$O_H(t) = e^{iHt} O_S e^{-iHt}, \quad \frac{dO_H}{dt} = i[H, O_H].$$

With $[H, a_{\vec{p}}] = -\omega_{\vec{p}} a_{\vec{p}}$ and $[H, a_{\vec{p}}^\dagger] = \omega_{\vec{p}} a_{\vec{p}}^\dagger$,

$$a_{\vec{p}}(t) = e^{-i\omega_{\vec{p}}t} a_{\vec{p}}, \quad a_{\vec{p}}^\dagger(t) = e^{i\omega_{\vec{p}}t} a_{\vec{p}}^\dagger,$$

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} (a_{\vec{p}} e^{-ip \cdot x} + a_{\vec{p}}^\dagger e^{ip \cdot x}) \Big|_{p^0 = E_{\vec{p}} = \omega_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}},$$

where $p^0 = E_{\vec{p}}$ is the **mass-shell condition**. Note that $e^{-ip \cdot x} = e^{-ip^0 t} e^{i\vec{p} \cdot \vec{x}}$ (the $e^{+ip^0 t}$ terms are the negative-frequency states). The equal-time commutation relation is

$$[\phi(\vec{x}, t), \pi(\vec{y}, t)] = i\delta^{(3)}(\vec{x} - \vec{y}),$$

and

$$\phi(x) = e^{i(Ht - P_i x^i)} \phi(0) e^{-i(Ht - P_i x^i)}, \quad P_\mu = (H, P_j).$$

Causality. In order to have a *causal* theory,

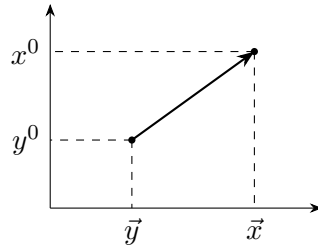
$$[\phi(x), \phi(y)] = 0 \quad \forall (x - y)^2 < 0 \text{ (spacelike)}.$$

Define

$$\Delta(x - y) := \langle 0 | [\phi(x), \phi(y)] | 0 \rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} (e^{-ip \cdot (x-y)} - e^{ip \cdot (x-y)}).$$

For spacelike separations one can choose a frame with $x^0 = y^0$, where the two terms cancel ($\vec{p} \rightarrow -\vec{p}$), so $\Delta = 0$; for timelike separations $(t, \vec{0})$ it behaves as $e^{-imt} - e^{imt} \neq 0$ (stationary phase approximation). Causality is obeyed thanks to the existence of particle–antiparticle pairs.

Propagation amplitude. Prepare a particle at the spacetime point y : what is the amplitude of finding it at x ?



$$\begin{aligned} D(x - y) &:= \langle 0 | \phi(x) \phi(y) | 0 \rangle = \int \frac{d^3p}{(2\pi)^3} \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{4E_{\vec{p}}E_{\vec{q}}}} \langle 0 | a_{\vec{p}} a_{\vec{q}}^\dagger | 0 \rangle e^{-ip \cdot x} e^{iq \cdot y} \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-ip \cdot (x-y)}. \end{aligned}$$

For timelike separations $D \sim e^{-imt}$; for spacelike separations $D \sim e^{-m|\vec{x}-\vec{y}|} \neq 0$: there is leakage outside the light cone. However, $\Delta(x - y) = D(x - y) - D(y - x)$ vanishes there.

4 Propagators and Interacting Scalar Fields

4.1 The Feynman propagator

$$\Delta_F(x-y) := \langle 0 | T \{ \phi(x) \phi(y) \} | 0 \rangle,$$

where the **time-ordering operator** moves to the left all operators evaluated at later times:

$$T \{ \phi(x) \phi(y) \} = \begin{cases} \phi(x) \phi(y), & x^0 > y^0, \\ \phi(y) \phi(x), & x^0 < y^0. \end{cases}$$

Hence

$$\begin{aligned} \Delta_F(x-y) &= \theta(x^0 - y^0) D(x-y) + \theta(y^0 - x^0) D(y-x) \\ &= \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} \left[\theta(x^0 - y^0) e^{-ip \cdot (x-y)} + \theta(y^0 - x^0) e^{ip \cdot (x-y)} \right] : \end{aligned}$$

a particle propagating from y to x or from x to y (the first bit of Feynman diagrams). Equivalently, integrating for the first time over p^0 ,

$$\Delta_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)}, \quad \tilde{\Delta}_F(p) = \frac{i}{p^2 - m^2 + i\epsilon},$$

the **Feynman propagator in momentum space**.

Proof. $p^2 - m^2 = (p^0)^2 - E_{\vec{p}}^2 = (p^0 + E_{\vec{p}})(p^0 - E_{\vec{p}})$, so the integrand has two poles, displaced off the real axis by the $i\epsilon$ prescription.

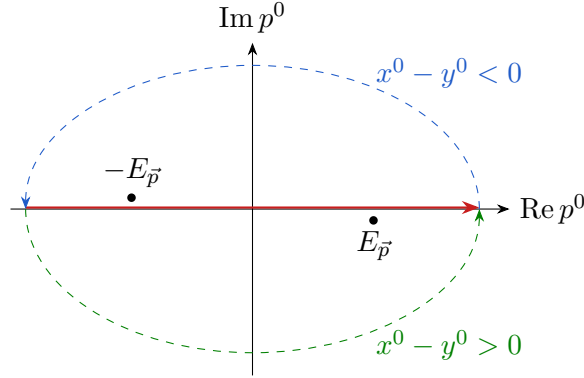


Figure 4.1: Poles of the Feynman propagator and the contours used for $x^0 > y^0$ (closing below) and $x^0 < y^0$ (closing above).

By the residue theorem, for $x^0 - y^0 > 0$ (closing clockwise below),

$$\int \frac{dp^0}{2\pi} \frac{i}{p^2 - m^2} e^{-ip^0(x^0 - y^0)} = \frac{1}{2E_{\vec{p}}} e^{-iE_{\vec{p}}(x^0 - y^0)},$$

and for $x^0 - y^0 < 0$ (closing above) one picks up the pole at $-E_{\vec{p}}$, giving $\frac{1}{2E_{\vec{p}}} e^{iE_{\vec{p}}(x^0 - y^0)}$. Combining with the spatial integral (and $\vec{p} \rightarrow -\vec{p}$ in the second term) reproduces the expression above. ■

4.2 Pictures

- Schrödinger picture: operators are time independent, states are time dependent, $|\psi\rangle = |\psi(t)\rangle$.

- Heisenberg picture: operators are time dependent, $O = O(t)$; states are not.
- Interaction picture: $H = H_0 + H_1$, with H_0 the free theory and H_1 a small perturbation; both operators and states depend on time.

In the interaction picture,

$$O_I(t) = e^{iH_0 t} O e^{-iH_0 t} \implies \frac{dO_I}{dt} = i[H_0, O_I], \quad |\psi_I(t)\rangle := e^{iH_0 t} |\psi(t)\rangle.$$

Expectation values agree, $\langle \psi(t) | O | \psi(t) \rangle = \langle \psi_I(t) | O_I(t) | \psi_I(t) \rangle$, and

$$\begin{aligned} i\partial_t |\psi_I(t)\rangle &= e^{iH_0 t} (-H_0 + H) e^{-iH_0 t} |\psi_I(t)\rangle \implies \\ i\partial_t |\psi_I(t)\rangle &= H_I(t) |\psi_I(t)\rangle, \quad H_I(t) := e^{iH_0 t} H_1 e^{-iH_0 t}. \end{aligned}$$

4.3 Dyson formula

With $|\psi_I(t)\rangle = U(t, t_0) |\psi_I(t_0)\rangle$ (H_I at different times need not commute),

$$i \frac{d}{dt} U(t, t_0) = H_I(t) U(t, t_0).$$

Solving perturbatively:

- order 0: $i \frac{dU^{(0)}}{dt} = 0 \Rightarrow U^{(0)} = \mathbb{1}$;
- order 1: $i \frac{dU^{(1)}}{dt} = H_I(t) U^{(0)} \Rightarrow U^{(1)} = -i \int_{t_0}^t dt' H_I(t')$;
- order 2: $i \frac{dU^{(2)}}{dt} = H_I(t) U^{(1)} \Rightarrow U^{(2)} = - \int_{t_0}^t d\tilde{t} \int_{t_0}^{\tilde{t}} dt' H_I(\tilde{t}) H_I(t')$,

a multiple integral with nested boundaries. Relabelling the integration region (the triangle $t_0 < t_2 < t_1 < t$),

$$U^{(2)} = - \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H_I(t_1) H_I(t_2) = -\frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 T\{H_I(t_1) H_I(t_2)\},$$

and in general

$$U^{(n)} = \frac{1}{n!} T \left\{ \left(-i \int_{t_0}^t dt' H_I(t') \right)^n \right\} \implies U(t, t_0) = T \exp \left(-i \int_{t_0}^t dt' H_I(t') \right),$$

the **Dyson formula**.

4.4 Scattering processes

We want scattering cross sections, matrices and amplitudes. $\langle f | S | i \rangle$ is the amplitude for the state $|i\rangle$ ($t \rightarrow -\infty$) to evolve into the final state $|f\rangle$ ($t \rightarrow +\infty$) via the unitary **S-matrix**, which describes the interaction:

$$\langle f | S | i \rangle := \lim_{\substack{t_0 \rightarrow -\infty \\ t \rightarrow +\infty}} \langle f | U(t, t_0) | i \rangle, \quad S = T \exp \left(-i \int_{-\infty}^{\infty} dt H_I(t) \right).$$

Writing $S = \mathbb{1} + i\mathcal{T}$,

$$\langle f | i\mathcal{T} | i \rangle = (2\pi)^4 \delta^{(4)}(p_f - p_i) i\mathcal{A}.$$

4.5 Wick's theorem

With $N\{\cdots\} := :(\cdots):$ and the **contraction** $\overline{\phi_1\phi_2} := \Delta_F(x_1 - x_2)$, $\phi_i := \phi(x_i)$,

$$T\{\phi_1\phi_2\cdots\phi_n\} = N\{\phi_1\phi_2\cdots\phi_n + \text{all possible contractions}\}.$$

For example,

$$T\{\phi_1\phi_2\phi_3\phi_4\} = N\{\phi_1\phi_2\phi_3\phi_4 + \overline{\phi_1\phi_2}\phi_3\phi_4 + \overline{\phi_1\phi_3}\phi_2\phi_4 + \cdots + \overline{\phi_1\phi_2}\phi_3\phi_4 + \overline{\phi_1\phi_3}\phi_2\phi_4 + \overline{\phi_1\phi_4}\phi_2\phi_3 + \overline{\phi_2\phi_3}\phi_1\phi_4\}.$$

Since the vacuum expectation value of any normal-ordered product vanishes,

$$\langle 0|T\{\phi_1\cdots\phi_n\}|0\rangle = \text{all possible full contractions}.$$

For $n = 4$,

$$\begin{aligned} \langle 0|T\{\phi_1\phi_2\phi_3\phi_4\}|0\rangle &= \Delta_F(x_1 - x_2)\Delta_F(x_3 - x_4) \\ &\quad + \Delta_F(x_1 - x_3)\Delta_F(x_2 - x_4) + \Delta_F(x_1 - x_4)\Delta_F(x_2 - x_3), \end{aligned}$$

represented by the three diagrams pairing the four points.

Proof for two fields. Split $\phi(x) = \phi^+(x) + \phi^-(x)$ with

$$\phi^+(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} a_{\vec{p}} e^{-ip \cdot x}, \quad \phi^-(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} a_{\vec{p}}^\dagger e^{ip \cdot x}.$$

Then

$$\begin{aligned} N\{\phi_1\phi_2\} &= \phi_1^+\phi_2^+ + \phi_1^-\phi_2^- + \phi_1^-\phi_2^+ + \phi_2^-\phi_1^+, \\ T\{\phi_1\phi_2\} &= \phi_1^+\phi_2^+ + \phi_1^-\phi_2^- + \theta(t_1 - t_2)[\phi_1^+\phi_2^- + \phi_1^-\phi_2^+] + \theta(t_2 - t_1)[\phi_2^+\phi_1^- + \phi_2^-\phi_1^+], \end{aligned}$$

so

$$T\{\phi_1\phi_2\} - N\{\phi_1\phi_2\} = \theta(t_1 - t_2)[\phi_1^+, \phi_2^-] + \theta(t_2 - t_1)[\phi_2^+, \phi_1^-] = \Delta_F(x_1 - x_2),$$

a c-number. ■

5 Interactions and Feynman Diagrams

5.1 Review

$$\begin{aligned} H_I &= e^{iH_0t} H_1 e^{-iH_0t}, \quad |\psi_I(t)\rangle = U_I(t, t_0) |\psi_I(t_0)\rangle, \\ i \frac{dU_I}{dt} &= H_I U_I \implies U_I(t, t_0) = T \exp \left(-i \int_{t_0}^t dt' H_I(t') \right), \end{aligned}$$

$$\langle f|S|i\rangle = \lim_{\substack{t_+ \rightarrow +\infty \\ t_- \rightarrow -\infty}} \langle f|U(t_+, t_-)|i\rangle, \quad S = \lim_{\substack{t_+ \rightarrow +\infty \\ t_- \rightarrow -\infty}} T \exp \left(-i \int_{t_-}^{t_+} dt H_I(t) \right).$$

To evaluate $\langle f|T\{\phi(x_1)\cdots\phi(x_n)\}|i\rangle$, Wick's theorem will be useful.

5.2 Interaction terms and dimensional analysis

Interactions can be added, in particular

$$\Delta\mathcal{L} = \mathcal{L}_{\text{int}} = - \sum_{n \geq 3} \frac{\lambda_n}{n!} \phi^n.$$

For a given energy only a finite number of these terms are relevant. With $\hbar = c = 1$ the action is dimensionless, $[S] = 0$ in mass dimensions ($[m] = 1$), so $[\mathcal{L}] = 4$; since $[\partial_\mu] = 1$, $[\phi] = 1$ and

$$[\lambda_n] = 4 - n.$$

It is convenient to use dimensionless couplings, $\tilde{\lambda}_n = \lambda_n/E^{4-n}$, where E is the relevant energy scale.

- $n = 3$: the ϕ^3 term is negligible at large energies but important at low energies: **relevant** term.
- $n = 4$: ϕ^4 matters at all energies: **marginal** term.
- $n > 4$: ϕ^n is important only at high energy scales: **irrelevant** term.

ϕ^4 theory.

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4.$$

Yukawa theory (a toy-model approximation of QED):

$$\mathcal{L}_Y = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \partial_\mu \psi^* \partial^\mu \psi - M^2 \psi^* \psi - g \psi^* \psi \phi.$$

5.3 Scattering in ϕ^4 theory

With $H_1 = \frac{\lambda}{4!} \phi^4$ and

$$|i\rangle = \sqrt{4E_{\vec{p}_1} E_{\vec{p}_2}} a_{\vec{p}_1}^\dagger a_{\vec{p}_2}^\dagger |0\rangle = |\vec{p}_1 \vec{p}_2\rangle, \quad |f\rangle = \sqrt{4E_{\vec{p}_3} E_{\vec{p}_4}} a_{\vec{p}_3}^\dagger a_{\vec{p}_4}^\dagger |0\rangle = |\vec{p}_3 \vec{p}_4\rangle,$$

the scattering amplitude is

$$\begin{aligned} \langle f | S | i \rangle &= \langle f | \lim T \left\{ \exp \left(-i \int dt d^3x \mathcal{H}_1 \right) \right\} | i \rangle \\ &= \langle \vec{p}_3 \vec{p}_4 | \vec{p}_1 \vec{p}_2 \rangle - \frac{i\lambda}{4!} \int d^4x \langle \vec{p}_3 \vec{p}_4 | T \{ \phi^4(x) \} | \vec{p}_1 \vec{p}_2 \rangle \\ &\quad - \frac{1}{2} \left(\frac{\lambda}{4!} \right)^2 \int d^4x d^4y \langle \vec{p}_3 \vec{p}_4 | T \{ \phi^4(x) \phi^4(y) \} | \vec{p}_1 \vec{p}_2 \rangle + \dots \end{aligned}$$

Zeroth order.

$$\begin{aligned} \langle \vec{p}_3 \vec{p}_4 | \vec{p}_1 \vec{p}_2 \rangle &= 4 \sqrt{E_1 E_2 E_3 E_4} \langle 0 | a_{\vec{p}_3} a_{\vec{p}_4} a_{\vec{p}_1}^\dagger a_{\vec{p}_2}^\dagger | 0 \rangle \\ &= 4 E_{\vec{p}_1} E_{\vec{p}_2} (2\pi)^6 [\delta^{(3)}(\vec{p}_3 - \vec{p}_1) \delta^{(3)}(\vec{p}_4 - \vec{p}_2) \\ &\quad + \delta^{(3)}(\vec{p}_3 - \vec{p}_2) \delta^{(3)}(\vec{p}_4 - \vec{p}_1)] : \end{aligned}$$

the two particles pass by without interacting.

First order. By Wick's theorem,

$$\langle \vec{p}_3 \vec{p}_4 | T \{ \phi_1 \phi_2 \phi_3 \phi_4 \} | \vec{p}_1 \vec{p}_2 \rangle = \langle \vec{p}_3 \vec{p}_4 | N \{ \phi_1 \phi_2 \phi_3 \phi_4 + \text{all possible contractions} \} | \vec{p}_1 \vec{p}_2 \rangle,$$

where uncontracted fields act on the external states:

$$\phi^+(x) |\vec{p}\rangle = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{k}}}} a_{\vec{k}} e^{-ik \cdot x} \sqrt{2E_{\vec{p}}} a_{\vec{p}}^\dagger |0\rangle = e^{-ip \cdot x} |0\rangle.$$

For $\phi^4(x)$ there are three kinds of terms (all at the same point x):

(A) two internal contractions: a **vacuum bubble** times the free propagation,

$$S_1^A = -\frac{i\lambda}{4!} \cdot 3 \langle \vec{p}_3 \vec{p}_4 | \vec{p}_1 \vec{p}_2 \rangle \int d^4x \Delta_F^2(x-x);$$

(B) one internal contraction (a loop on a line): these are **disconnected diagrams**, where a particle goes from the initial to the final state along the same line,

$$S_1^B = -\frac{i\lambda}{4!} \langle \vec{p}_4 | \vec{p}_2 \rangle \int d^4x \Delta_F(x-x) e^{i(p_3-p_1) \cdot x} + 3 \text{ other terms};$$

(C) all four fields contracted with the external states, with a combinatorial factor $4!$:

$$S_1^C = 4! \left(-\frac{i\lambda}{4!} \right) \int d^4x e^{i(p_3+p_4-p_1-p_2) \cdot x} = -i\lambda (2\pi)^4 \delta^{(4)}(p_3+p_4-p_1-p_2).$$

Hence the **tree-level scattering amplitude** is $i\mathcal{A} = -i\lambda$. The vacuum bubbles are eliminated by the normalisation with respect to the vacuum $|\Omega\rangle$ of the interacting theory (see below), and disconnected diagrams do not contribute to scattering. Writing $S = \mathbb{K} + i\mathcal{T}$,

$$\langle f | i\mathcal{T} | i \rangle = (2\pi)^4 \delta^{(4)}(p_f - p_i) i\mathcal{A}.$$

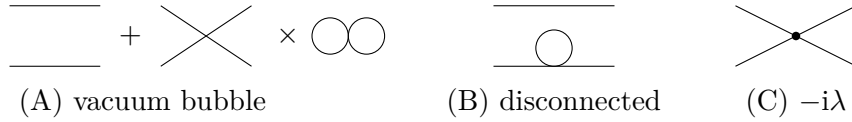


Figure 5.1: First-order contributions to $2 \rightarrow 2$ scattering in ϕ^4 theory.

6 Interacting Scalar Field and Feynman Diagrams

(Peskin–Schroeder 4.2, 4.4, 4.6.)

6.1 Second order

Since we are only interested in full contractions with the external fields,

$$S_2 = \frac{1}{2} \left(-\frac{i\lambda}{4!} \right)^2 \int d^4x d^4y \langle \vec{p}_3 \vec{p}_4 | \phi_x \phi_x \phi_x \phi_x \phi_y \phi_y \phi_y \phi_y | \vec{p}_1 \vec{p}_2 \rangle$$

with the external momenta p_1, p_2 contracted at y , p_3, p_4 at x , and two propagators between x and y . Counting: $12 \cdot 12$ ways to attach the external lines, 2 ways to join the remaining fields, and 2 for $x \leftrightarrow y$, i.e. $576 = (4!)^2$. Hence

$$S_2 = \frac{(-i\lambda)^2}{2} \int d^4x d^4y \Delta_F(x-y) \Delta_F(x-y) e^{i(p_3+p_4) \cdot x} e^{-i(p_1+p_2) \cdot y}.$$

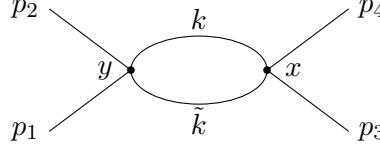


Figure 6.1: One-loop (s -channel) contribution in ϕ^4 theory.

In momentum space,

$$S_2 = \frac{(-i\lambda)^2}{2} \int d^4x d^4y e^{-i(p_1+p_2)\cdot y} e^{i(p_3+p_4)\cdot x} \int \frac{d^4k}{(2\pi)^4} \frac{ie^{-ik\cdot(x-y)}}{k^2 - m^2 + i\epsilon} \int \frac{d^4\tilde{k}}{(2\pi)^4} \frac{ie^{-i\tilde{k}\cdot(x-y)}}{\tilde{k}^2 - m^2 + i\epsilon},$$

which gives

$$S_2 = (2\pi)^4 \delta^{(4)}(p_3 + p_4 - p_1 - p_2) \frac{(-i\lambda)^2}{2} \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(p_1 + p_2 - k)^2 - m^2 + i\epsilon},$$

$$i\mathcal{A}_2 = \frac{(-i\lambda)^2}{2} \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(p_1 + p_2 - k)^2 - m^2 + i\epsilon}.$$

For large k , $\mathcal{A}_2 \sim \int^\Lambda d^4k k^{-4} \sim \ln \Lambda$: a logarithmic divergence.

6.2 Feynman diagrams and rules

Steps (position space or momentum space):

1. Draw all (relevant) diagrams: draw an external line for each particle in $|i\rangle$ and $|f\rangle$, and join the external lines using the vertices given by H_I .
2. Assign an amplitude:
 - each vertex contributes $-i\lambda$ (for this theory), together with an integration over all spacetime;
 - each internal line ($x \rightarrow y$) contributes $\Delta_F(x - y)$ (in momentum space $\tilde{\Delta}_F(k)$, with an integration over unconstrained internal momenta);
 - each external line contributes $e^{-ip\cdot x}$ (incoming) or $e^{ip\cdot x}$ (outgoing) in position space, and 1 in momentum space;
 - divide by the symmetry factor;
 - impose momentum conservation at each vertex.

6.3 Why do we discard disconnected diagrams and vacuum bubbles?

(Peskin–Schroeder 4.2, 4.6.) The amplitude

$$\langle f | (S - \mathbb{K}) | i \rangle = \lim_{t_\pm \rightarrow \pm\infty} \left(\langle f | T \left\{ e^{-i \int d^4x \mathcal{H}_I} \right\} | i \rangle \right)_0$$

is computed with free states and the free vacuum, keeping only external lines fully connected. With $H = H_0 + H_I$, the interaction-picture field is the free field, $\phi_I(x) = e^{iH_0 t} \phi(\vec{x}) e^{-iH_0 t}$, related to the Heisenberg field by $\phi(x) = U^{-1}(t, 0) \phi_I(x) U(t, 0)$, and $\phi_I(x) |\vec{p}\rangle = e^{-ip\cdot x} |0\rangle$ involves the free vacuum, $H_0 |0\rangle = 0$. It should be the vacuum $|\Omega\rangle$ of the interacting theory, $H |\Omega\rangle = E_0 |\Omega\rangle$: the two are different, $|0\rangle \neq |\Omega\rangle$, but they overlap, $\langle \Omega | 0 \rangle \neq 0$. Evolving adiabatically,

$$|\Omega\rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} (e^{-iE_0 T} \langle \Omega | 0 \rangle)^{-1} e^{-iHT} |0\rangle, \quad |\vec{p}_1 \vec{p}_2\rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} e^{-iHT} |\vec{p}_1 \vec{p}_2\rangle_0,$$

and

$$\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = \lim_{T \rightarrow \infty (1-i\epsilon)} \frac{\langle 0 | T \{ \phi_I(x) \phi_I(y) \exp(-i \int_{-T}^T dt H_I(t)) \} | 0 \rangle}{\langle 0 | T \{ \exp(-i \int_{-T}^T dt H_I(t)) \} | 0 \rangle},$$

where the denominator, made of vacuum bubble diagrams (the “particle sea”), cancels the vacuum bubbles of the numerator.

7 Interacting Scalar Fields and Fermions

7.1 Scalar Yukawa theory

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu \phi)^2 - \frac{m^2}{2}\phi^2}_{\text{“photon-like”}} + \underbrace{\partial_\mu \psi^* \partial^\mu \psi - M^2 \psi^* \psi}_{\text{“electron-like”}} - \underbrace{g \psi^* \psi \phi}_{\text{vertex}}.$$

In this case the coupling has dimensions of energy, $[g] = 1$; in the fermion version (the useful one) g is dimensionless. In diagrams, ϕ is drawn as a dashed line and ψ as a solid line, with arrows indicating the flow of charge.

“Electron” scattering $\psi\psi \rightarrow \psi\psi$. At order g^2 , with $|i\rangle = \sqrt{4E_{\vec{p}_1}E_{\vec{p}_2}} b_{\vec{p}_1}^\dagger b_{\vec{p}_2}^\dagger |0\rangle = |\vec{p}_1 \vec{p}_2\rangle$ and $|f\rangle = |\vec{p}_3 \vec{p}_4\rangle$,

$$\langle f | S | i \rangle = \frac{(-ig)^2}{2} \int d^4x d^4y \langle \vec{p}_3 \vec{p}_4 | T \{ (\psi^* \psi \phi)_x (\psi^* \psi \phi)_y \} | \vec{p}_1 \vec{p}_2 \rangle.$$

Contracting $\phi_x \phi_y$ and the ψ 's with the external states (plus $x \leftrightarrow y$, which cancels the $\frac{1}{2}$), with $\psi(x) |\vec{p}\rangle = e^{-ip \cdot x} |0\rangle$, $\langle \vec{p} | \psi^*(x) = \langle 0 | e^{ip \cdot x}$,

$$\begin{aligned} \langle f | S | i \rangle &= (-ig)^2 \int d^4x d^4y e^{i(p_3-p_1) \cdot x} e^{i(p_4-p_2) \cdot y} \int \frac{d^4q}{(2\pi)^4} \frac{ie^{-iq \cdot (x-y)}}{q^2 - m^2} \\ &= (-ig)^2 \int \frac{d^4q}{(2\pi)^4} \frac{i}{q^2 - m^2} (2\pi)^4 \delta^{(4)}(p_3 - p_1 - q) (2\pi)^4 \delta^{(4)}(p_4 - p_2 + q) \\ &= i(-ig)^2 \frac{1}{(p_1 - p_3)^2 - m^2} (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4), \end{aligned}$$

$$\mathcal{A}_t = (-ig)^2 \frac{1}{(p_1 - p_3)^2 - m^2} \quad (t\text{-channel}), \quad \mathcal{A}_u = (-ig)^2 \frac{1}{(p_1 - p_4)^2 - m^2} \quad (u\text{-channel}),$$

the second coming from the other contraction ($p_3 \leftrightarrow p_4$).

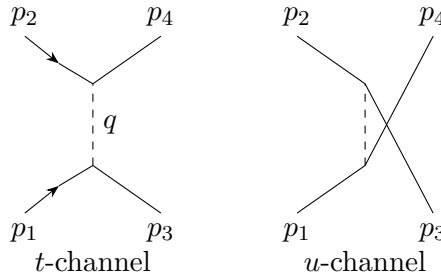


Figure 7.1: Order- g^2 diagrams for $\psi\psi \rightarrow \psi\psi$ in scalar Yukawa theory.

7.2 Mandelstam variables (for $2 \rightarrow 2$)

$$s := (p_1 + p_2)^2, \quad t := (p_1 - p_3)^2, \quad u := (p_1 - p_4)^2,$$

where s is the centre-of-mass energy squared and t, u measure the momentum exchange (and include the scattering angle). They satisfy

$$\sum_i m_i^2 = s + t + u.$$

Why is $i\epsilon$ not included in these propagators? Before, it was needed because of the on-shell condition; now the particle in between is virtual and does not necessarily satisfy the on-shell condition. Be careful with the s -channel, where $s = m^2$ is possible: add $i\epsilon$.

Feynman rules for scalar Yukawa theory.

- each vertex: $-ig$;
- each external line: 1;
- internal lines: $\frac{i}{k^2 - m^2 + i\epsilon}$ (dashed), $\frac{i}{k^2 - M^2 + i\epsilon}$ (solid);
- momentum conservation at each vertex;
- integrate over all unconstrained internal momenta.

Some diagrams.

- (a) Scattering of a ψ particle off a ϕ particle ($\psi\phi \rightarrow \psi\phi$), exchanging a ψ (two diagrams):

$$\mathcal{A} = (-ig)^2 \left(\frac{1}{t - M^2} + \frac{1}{u - M^2} \right),$$

with t and u defined by the momentum q flowing through the internal line in each diagram.

- (b) $\psi\bar{\psi} \rightarrow \psi\bar{\psi}$ (“electron–positron” scattering): t -channel and s -channel,

$$\mathcal{A} = (-ig)^2 \left(\frac{1}{t - m^2} + \frac{1}{s - m^2 + i\epsilon} \right).$$

The s -channel shows resonances, which is how new particles are found.

- (c) $\phi\phi \rightarrow \phi\phi$ (“photon” scattering): at order g^4 the simplest diagram involves a loop,

$$\mathcal{A} = (-ig)^4 \int \frac{d^4k}{(2\pi)^4} \frac{1}{D_0 D_1 D_{12} D_3},$$

with $D_0 = k^2 - M^2 + i\epsilon$, $D_1 = (k - p_1)^2 - M^2 + i\epsilon$, $D_{12} = (k - p_1 - p_2)^2 - M^2 + i\epsilon$ and $D_3 = (k - p_3)^2 - M^2 + i\epsilon$.

7.3 Relation to observables

Decay rates, cross sections and correlations. The Lorentz-invariant phase-space element for N final particles is

$$d\Pi := (2\pi)^4 \delta^{(4)}(p_f - p_i) \prod_{n=1}^N \frac{d^3p_n}{(2\pi)^3} \frac{1}{2E_{\vec{p}_n}}.$$

Decay rate of a particle of mass M (in its rest frame):

$$\Gamma_0 = \frac{1}{2M} \sum_f \int |\mathcal{A}|^2 d\Pi,$$

where $\int |\mathcal{A}|^2 d\Pi$ is Lorentz invariant but the prefactor must be corrected in a moving frame ($E = \gamma M$, $\gamma = 1/\sqrt{1 - v^2}$): $\Gamma = \Gamma_0/\gamma$. The half-life $\tau_0 = \Gamma_0^{-1}$ becomes $\tau = \gamma\tau_0$.

Cross section. The rate of scattering is $R = \sigma \mathcal{L}$, where the luminosity $\mathcal{L}$ is the flux of particles per unit area per unit time and σ the scattering cross section. For $2 \rightarrow N$,

$$\sigma = \frac{1}{4I} \sum_f \int |\mathcal{A}|^2 d\Pi, \quad I = \sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} \quad (\text{flux factor}).$$

I is frame dependent, and so is σ ; it is usually calculated in the centre-of-mass frame. For $2 \rightarrow 2$ in the centre-of-mass frame, only the solid-angle integration is left:

$$d\Pi = \frac{1}{(2\pi)^2} \frac{|\vec{p}'|}{4\sqrt{s}} d\Omega, \quad s = (p_1 + p_2)^2 = E_{\text{cm}}^2,$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\vec{p}'|}{|\vec{p}|} |\mathcal{A}|^2 \quad (\vec{p} \text{ incoming, } \vec{p}' \text{ outgoing}), \quad \left(\frac{d\sigma}{d\Omega} \right)_{\text{cm}}^{\text{elastic}} = \frac{|\mathcal{A}|^2}{64\pi^2 s}.$$

Peaks in the cross section signal particles.

8 Fermions

8.1 Groups and representations

- $SU(2)$: the group of 2×2 unitary matrices with determinant 1; the Pauli matrices σ (times i) form a basis of its algebra.
- Lorentz group $SO(3, 1)$; rotation group $SO(3)$: $R^\top R = \mathbb{1}$, $\det R = 1$. $SO(3)$ is a subgroup of $SO(3, 1)$.
- V^μ is an irreducible representation of $SO(3, 1)$ but reducible under $SO(3)$ (scalar $\oplus$ vector).

Particles are identified with irreducible representations of the Poincaré group (Lorentz transformations and translations).

- A **group** is a set of elements $\{g_i\}$ with a composition rule $\cdot$ such that $g_i \cdot g_j = g_k$ (plus associativity, identity and inverses).
- A **representation** of a group is a map (embedding) of the group elements g_i into operators that act on a vector space, preserving the composition rule. For finite-dimensional representations the elements can always be mapped to matrices.
- A representation is **irreducible** if there is no proper subspace that transforms among itself.

8.2 The Lorentz group $SO(3, 1)$

A set of objects ϕ^i , $i = 1, \dots, n$, transforms in a representation R of dimension n if under a Lorentz transformation

$$\phi^i \rightarrow \left[e^{-\frac{i}{2} \omega_{\mu\nu} J_R^{\mu\nu}} \right]^i_j \phi^j,$$

where the $J_R^{\mu\nu}$ ($n \times n$ matrices) are the Lorentz generators in the representation R , i.e. $\delta\phi^i = -\frac{i}{2} \omega_{\mu\nu} (J_R^{\mu\nu})^i_j \phi^j$; (μ, ν) label the generators and (i, j) are matrix indices.

- Scalar field: $\phi'(x') = \phi(x)$, $i, j \in \{1\}$, $J_{\text{scalar}}^{\mu\nu} = 0$: the trivial representation (the change of $\phi(x)$ comes only from the argument, $\delta\phi = -\omega^\mu{}_\nu x^\nu \partial_\mu \phi$; with $\Lambda = e^{-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}}$ and $J^{\mu\nu} = -J^{\nu\mu}$ there are 6 independent generators).
- Four-vector V^μ : the indices i, j are themselves Lorentz indices, and

$$(J^{\mu\nu})^\rho{}_\sigma = i(\eta^{\mu\rho} \delta_\sigma^\nu - \eta^{\nu\rho} \delta_\sigma^\mu).$$

The generators satisfy the Lie algebra of $SO(3, 1)$:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}).$$

Writing $J^i = \frac{1}{2}\varepsilon^{ijk} J^{jk}$ and $K^i = J^{i0}$, so that $\Lambda = e^{i\vec{\theta}\cdot\vec{J} + i\vec{\beta}\cdot\vec{K}}$,

$$\begin{aligned} [J^i, J^j] &= i\varepsilon^{ijk} J^k & (SO(3) \simeq SU(2)), \\ [J^i, K^j] &= i\varepsilon^{ijk} K^k & (\vec{K} \text{ is a vector}), \\ [K^i, K^j] &= -i\varepsilon^{ijk} J^k. \end{aligned}$$

8.3 $SU(2)$ and spinors

$SU(2)$ is the group of unitary matrices with determinant 1; it is periodic only under rotations by 4π ($SU(2)$ double-covers $SO(3)$). Its representations are labelled by $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, and the spin- j representation has $2j + 1$ dimensions. The spinorial (fundamental) representation $j = \frac{1}{2}$ is two dimensional,

$$J^i = \frac{\sigma^i}{2}, \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma^i \sigma^j = \delta^{ij} + i\varepsilon^{ijk} \sigma^k.$$

All other representations are built from tensor products of spin- $\frac{1}{2}$ representations, e.g. $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ (dimensions $2 \times 2 = 1 + 3$).

Combining the $SO(3, 1)$ generators as

$$\vec{J}_\pm = \frac{\vec{J} \pm i\vec{K}}{2} \implies [J_\pm^i, J_\pm^j] = i\varepsilon^{ijk} J_\pm^k, \quad [J_+^i, J_-^j] = 0 :$$

the Lorentz algebra is composed of two (commuting) copies of the $SU(2)$ algebra. Representations of the Lorentz group can therefore be labelled by (j_+, j_-) , with dimension $(2j_+ + 1)(2j_- + 1)$, rotation generators $\vec{J} = \vec{J}_+ + \vec{J}_-$, and spins from $|j_+ - j_-|$ to $j_+ + j_-$.

Spin- $\frac{1}{2}$ particles are associated with $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$, the **Weyl spinors**: one has $\vec{J} = \vec{\sigma}/2$, $\vec{K} = i\vec{\sigma}/2$ (left-handed spinor ψ_L) and the other $\vec{J} = \vec{\sigma}/2$, $\vec{K} = -i\vec{\sigma}/2$ (right-handed spinor ψ_R):

$$\begin{aligned} \psi_L &\rightarrow \Lambda_L \psi_L = e^{\frac{1}{2}(i\vec{\theta} - \vec{\beta})\cdot\vec{\sigma}} \psi_L, & \psi_R &\rightarrow \Lambda_R \psi_R = e^{\frac{1}{2}(i\vec{\theta} + \vec{\beta})\cdot\vec{\sigma}} \psi_R, \\ \delta\psi_L &= \frac{1}{2}(i\theta_i - \beta_i)\sigma^i \psi_L, & \delta\psi_R &= \frac{1}{2}(i\theta_i + \beta_i)\sigma^i \psi_R. \end{aligned}$$

8.4 Lorentz-invariant Lagrangian for Weyl and Dirac spinors

To have a Lorentz-invariant mass term one needs to combine ψ_L and ψ_R : $m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L)$. The kinetic terms are $i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L + i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R$, with $\sigma^\mu := (\mathbb{1}, \sigma^i)$, $\bar{\sigma}^\mu := (\mathbb{1}, -\sigma^i)$ ($\psi_L^\dagger \bar{\sigma}^\mu \psi_L$ is a four-vector). Hence

$$\mathcal{L} = i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L + i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R - m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L).$$

For $m \rightarrow 0$: $\sigma^\mu \partial_\mu \psi_R = 0$ and $\bar{\sigma}^\mu \partial_\mu \psi_L = 0$, i.e.

$$\frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|} \psi_R = \psi_R, \quad \frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|} \psi_L = -\psi_L :$$

ψ_R and ψ_L are eigenvectors of the **helicity operator** $h := \vec{\sigma} \cdot \vec{p}/|\vec{p}|$ with eigenvalues $+1$ and -1 .

Dirac equation. The four-component **Dirac spinor**, the gamma matrices and the Dirac adjoint are

$$\psi := \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad \gamma^\mu := \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \bar{\psi} := \psi^\dagger \gamma^0 = (\psi_R^\dagger, \psi_L^\dagger),$$

and

$$\mathcal{L}_D = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi = \bar{\psi}(i\cancel{\partial} - m)\psi \implies (i\gamma^\mu \partial_\mu - m)\psi = 0,$$

the **Dirac equation**. Applying $(-i\cancel{\partial} - m)$ to it yields the Klein–Gordon equation, $(\partial_\mu \partial^\mu + m^2)\psi = 0$. The conjugate momentum is $\pi_\psi = i\psi^\dagger$, so there are 4 degrees of freedom. $\bar{\psi}\psi$ is a scalar and $\bar{\psi}\gamma^\mu\psi$ a vector.

Lorentz transformation of Dirac spinors.

$$\Lambda_V = e^{i\theta_j J^j + i\beta_j K^j}, \quad \Lambda_S = \begin{pmatrix} \Lambda_L & 0 \\ 0 & \Lambda_R \end{pmatrix} = e^{-\frac{i}{2}\omega_{\rho\sigma} S^{\rho\sigma}}, \quad S^{\rho\sigma} := \frac{i}{4}[\gamma^\rho, \gamma^\sigma],$$

with $\omega_{ij} = -\varepsilon_{ijk}\theta_k$, $\omega_{0i} = -\beta_i$, and

$$S^{0i} = -\frac{i}{2} \begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}, \quad S^{ij} = \frac{1}{2} \varepsilon^{ijk} \begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}.$$

The field also receives a contribution from the change of coordinates:

$$\delta\psi^\alpha = -\omega^\mu{}_\nu x^\nu \partial_\mu \psi^\alpha - \frac{i}{2} \omega_{\rho\sigma} (S^{\rho\sigma})^\alpha{}_\beta \psi^\beta.$$

9 Fermions II

9.1 Noether currents

- Translations, $x'^\mu = x^\mu + c^\mu$, $\delta\psi = -c^\mu \partial_\mu \psi$:

$$T^{\mu\nu} = i\bar{\psi}\gamma^\mu \partial^\nu \psi, \quad E = \int d^3x T^{00} = \int d^3x \bar{\psi}(-i\gamma^i \partial_i + m)\psi.$$

- Lorentz, $x'^\mu = \Lambda^\mu{}_\nu x^\nu = (\delta^\mu_\nu + \omega^\mu{}_\nu)x^\nu$:

$$(\mathcal{J}^\mu)^{\rho\sigma} = (x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho}) + \bar{\psi}\gamma^\mu S^{\rho\sigma}\psi.$$

9.2 Quantisation of the Dirac field

Starting from $(i\gamma^\mu \partial_\mu - m)\psi = 0$, the ansatz $\psi(x) = u(p)e^{-ip \cdot x}$ (positive-frequency solutions) gives

$$(\gamma^\mu p_\mu - m)u(p) = 0 \implies \begin{pmatrix} -m & p \cdot \sigma \\ p \cdot \bar{\sigma} & -m \end{pmatrix} u(p) = 0.$$

Using $(p \cdot \sigma)(p \cdot \bar{\sigma}) = p^2 = m^2$,

$$u(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi \\ \sqrt{p \cdot \bar{\sigma}} \xi \end{pmatrix}, \quad \xi^\dagger \xi = 1.$$

Similarly the negative-frequency solutions $\psi(x) = v(p)e^{ip \cdot x}$ have

$$v(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \eta \\ -\sqrt{p \cdot \bar{\sigma}} \eta \end{pmatrix}, \quad \eta^\dagger \eta = 1.$$

In general the basis chosen is the eigenbasis of σ^3 : $\xi_\uparrow = \eta_\uparrow = (1, 0)^\top$, $\xi_\downarrow = \eta_\downarrow = (0, 1)^\top$.

- Rest frame, $p^\mu = (m, 0, 0, 0)$:

$$u_\uparrow = \sqrt{m} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_\downarrow = \sqrt{m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_\uparrow = \sqrt{m} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad v_\downarrow = \sqrt{m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

- Boost in the z direction:

$$u_\uparrow = \begin{pmatrix} \sqrt{E-p_3} \\ 0 \\ \sqrt{E+p_3} \\ 0 \end{pmatrix}, \quad u_\downarrow = \begin{pmatrix} 0 \\ \sqrt{E+p_3} \\ 0 \\ \sqrt{E-p_3} \end{pmatrix}, \quad v_\uparrow = \begin{pmatrix} \sqrt{E-p_3} \\ 0 \\ -\sqrt{E+p_3} \\ 0 \end{pmatrix}, \quad v_\downarrow = \begin{pmatrix} 0 \\ \sqrt{E+p_3} \\ 0 \\ -\sqrt{E-p_3} \end{pmatrix}.$$

- Massless limit $m \rightarrow 0$ ($p_3 = E$):

$$u_\uparrow = \sqrt{2E} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_\downarrow = \sqrt{2E} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_\uparrow = \sqrt{2E} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad v_\downarrow = \sqrt{2E} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

These are eigenstates of the helicity operator $h = \frac{i}{2}\varepsilon_{ijk}\hat{p}^i S^{jk} = \frac{1}{2}\hat{p}_i \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix}$: spin-up and spin-down Weyl spinors.

In general the helicity of a particle is frame dependent; for massless particles it is invariant.

Canonical anticommutation relations. With $\pi_\psi = i\psi^\dagger$, one could naively postulate $[\psi_\alpha(\vec{x}), \psi_\beta^\dagger(\vec{y})] = \delta^{(3)}(\vec{x} - \vec{y})\delta_{\alpha\beta}$, but this runs into trouble. The correct way to quantise ψ is to impose

$$\{\psi_\alpha(\vec{x}), \psi_\beta^\dagger(\vec{y})\} = \delta^{(3)}(\vec{x} - \vec{y})\delta_{\alpha\beta}.$$

Then

$$\begin{aligned} \psi(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1}^2 \left(b_{\vec{p}}^s u^s(\vec{p}) e^{i\vec{p}\cdot\vec{x}} + c_{\vec{p}}^{s\dagger} v^s(\vec{p}) e^{-i\vec{p}\cdot\vec{x}} \right), \\ \psi^\dagger(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1}^2 \left(b_{\vec{p}}^{s\dagger} u^{s\dagger}(\vec{p}) e^{-i\vec{p}\cdot\vec{x}} + c_{\vec{p}}^s v^{s\dagger}(\vec{p}) e^{i\vec{p}\cdot\vec{x}} \right), \end{aligned}$$

$$\{b_{\vec{p}}^s, b_{\vec{q}}^{r\dagger}\} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \delta^{sr}, \quad \{c_{\vec{p}}^s, c_{\vec{q}}^{r\dagger}\} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \delta^{sr}.$$

The Hamiltonian is

$$H = \int d^3x (\pi_\psi \dot{\psi} - \mathcal{L}) = \int d^3x \bar{\psi} (-i\gamma^i \partial_i + m) \psi,$$

$$:H: = \int \frac{d^3p}{(2\pi)^3} \sum_s E_{\vec{p}} (b_{\vec{p}}^{s\dagger} b_{\vec{p}}^s + c_{\vec{p}}^{s\dagger} c_{\vec{p}}^s), \quad H = :H: - (2\pi)^3 \delta^{(3)}(0) \dots,$$

where the vacuum energy has the opposite sign compared with the bosonic system; this could solve the problem of the infinite energy (supersymmetry). Also $[H, b_{\vec{p}}^{s\dagger}] = E_{\vec{p}} b_{\vec{p}}^{s\dagger}$ and $[H, c_{\vec{p}}^{s\dagger}] = E_{\vec{p}} c_{\vec{p}}^{s\dagger}$. (Next time: spin-statistics.)

9.3 Internal symmetries

- $\psi \rightarrow e^{i\alpha}\psi$ gives the **vector current** $j^\mu = \bar{\psi}\gamma^\mu\psi$, with $Q = \int d^3x \psi^\dagger\psi$ and

$$:Q: = \int \frac{d^3p}{(2\pi)^3} \sum_s [b_{\vec{p}}^{s\dagger} b_{\vec{p}}^s - c_{\vec{p}}^{s\dagger} c_{\vec{p}}^s] :$$

fermions and antifermions have opposite charge.

- For $m \rightarrow 0$, **chiral symmetry**: $\psi \rightarrow e^{i\alpha\gamma^5}\psi$ with $\gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix}$, giving the **axial current** $j_A^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi$, with $\partial_\mu j_A^\mu = 2im\bar{\psi}\gamma^5\psi = 0$ for $m = 0$.

9.4 Feynman propagator for fermions

In the Heisenberg picture,

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1}^2 \left(b_{\vec{p}}^s u^s(p) e^{-ip \cdot x} + c_{\vec{p}}^{s\dagger} v^s(p) e^{ip \cdot x} \right),$$

and

$$S_F(x-y) = \langle 0 | T \{ \psi(x) \bar{\psi}(y) \} | 0 \rangle, \quad T \{ \psi(x) \bar{\psi}(y) \} = \begin{cases} \psi(x) \bar{\psi}(y), & x^0 > y^0, \\ -\bar{\psi}(y) \psi(x), & x^0 < y^0. \end{cases}$$

Using $\sum_s u^s \bar{u}^s = \not{p} + m$ and $\sum_s v^s \bar{v}^s = \not{p} - m$,

$$\begin{aligned} \langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} (\not{p} + m) e^{-ip \cdot (x-y)}, \\ \langle 0 | \bar{\psi}(y) \psi(x) | 0 \rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} (\not{p} - m) e^{ip \cdot (x-y)}, \end{aligned}$$

$$S_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x-y)} = (i\not{\partial}_x + m) \Delta_F(x-y), \quad S_F(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}.$$

9.5 Yukawa theory with fermions

(Peskin-Schroeder 4.7.)

$$\mathcal{L}_Y = \frac{1}{2}(\partial\phi)^2 - \frac{\mu^2}{2}\phi^2 + \bar{\psi}(i\not{\partial} - m)\psi - g\phi\bar{\psi}\psi.$$

Feynman rules (momentum space):

- incoming fermion: $u^s(p)$; outgoing fermion: $\bar{u}^s(p)$; incoming antifermion: $\bar{v}^s(p)$; outgoing antifermion: $v^s(p)$;
- vertex: $-ig$, with momentum conservation;
- integrate over undetermined internal momenta;
- propagators: $\frac{i}{p^2 - \mu^2 + i\epsilon}$ (scalar), $\frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}$ (fermion);
- a minus sign for every closed fermion loop, for every open fermion line that contains an outgoing antifermion, and for every crossing of fermions.

Example: $\psi\psi \rightarrow \psi\psi$ at order g^2 . With $H_I = g\bar{\psi}\psi\phi$,

$$\langle \vec{p}_3 \vec{p}_4 | S | \vec{p}_1 \vec{p}_2 \rangle = \frac{(-ig)^2}{2!} \langle \vec{p}_3 \vec{p}_4 | T \left(\int d^4x \bar{\psi}_I \psi_I \phi_I(x) \int d^4y \bar{\psi}_I \psi_I \phi_I(y) \right) | \vec{p}_1 \vec{p}_2 \rangle,$$

with $\langle \vec{p}_3 \vec{p}_4 | \propto \langle 0 | b_{\vec{p}_4} b_{\vec{p}_3}$ and $|\vec{p}_1 \vec{p}_2\rangle \propto b_{\vec{p}_1}^\dagger b_{\vec{p}_2}^\dagger |0\rangle$. The factor $2!$ from $x \leftrightarrow y$ cancels the $\frac{1}{2!}$ of the Dyson formula (no further combinatorial factors). Two contractions contribute:

$$\begin{aligned} i\mathcal{A}_t &= \frac{ig^2}{(p_1 - p_3)^2 - \mu^2 + i\epsilon} [\bar{u}^{s_3}(p_3) u^{s_1}(p_1)] [\bar{u}^{s_4}(p_4) u^{s_2}(p_2)], \\ i\mathcal{A}_u &= \frac{-ig^2}{(p_1 - p_4)^2 - \mu^2 + i\epsilon} [\bar{u}^{s_4}(p_4) u^{s_1}(p_1)] [\bar{u}^{s_3}(p_3) u^{s_2}(p_2)], \end{aligned}$$

with a relative minus sign from the exchange of identical fermions.

Fermion loops. For the diagram with a closed fermion loop and four external scalars,

$$\begin{aligned} \sum_{abcd} \bar{\psi}_a(x_1) \psi_a(x_1) \bar{\psi}_b(x_2) \psi_b(x_2) \bar{\psi}_c(x_3) \psi_c(x_3) \bar{\psi}_d(x_4) \psi_d(x_4) \\ = -\text{Tr}[S_F(x_4 - x_1) S_F(x_1 - x_2) S_F(x_2 - x_3) S_F(x_3 - x_4)]. \end{aligned}$$

t -channel of $\psi\bar{\psi}$ annihilation into scalars.

$$i\mathcal{A}_t = \frac{-ig^2}{(p_1 - p_3)^2 - m^2 + i\epsilon} \bar{v}^{s_2}(p_2) (\not{p}_1 - \not{p}_3 + m) u^{s_1}(p_1).$$

Squared amplitudes. For $d\sigma/d\Omega \propto |\mathcal{A}|^2$,

$$\mathcal{A}\mathcal{A}^* = [\bar{u}(p)\Gamma_1 u(k)][\bar{u}(p)\Gamma_2 u(k)]^*, \quad [\bar{u}(p)\Gamma_2 u(k)]^* = \bar{u}(k)\gamma^0\Gamma_2^\dagger\gamma^0 u(p),$$

and summing over spins with $\sum_r u^r(p)\bar{u}^r(p) = \not{p} + m$,

$$\sum_{\text{spins}} [\bar{u}(p)\Gamma_1 u(k)][\bar{u}(p)\Gamma_2 u(k)]^* = \text{Tr}\{(\not{p} + m)\Gamma_1(\not{k} + m)\gamma^0\Gamma_2^\dagger\gamma^0\}.$$

9.6 Gauge fields

Maxwell Lagrangian (spin-one field):

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu,$$

invariant under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$ (**gauge invariance**). The equations of motion are

$$\square A_\mu - \partial_\mu \partial_\nu A^\nu = 0, \quad \text{i.e.} \quad -\nabla^2 A_0 - \partial_0 \partial_i A^i = 0, \quad \square A_i - \partial_i (\partial_0 A_0 - \partial_j A^j) = 0.$$

Fixing the gauge.

- Coulomb gauge, $\partial_i A_i = 0$ (not completely fixed): then $\nabla^2 A_0 = 0$, and with the residual freedom $A_0 \rightarrow A_0 + \partial_0 \alpha$ we set $A_0 = 0$, so $\square A_i = 0$: $\vec{A}$ has just two degrees of freedom.
- Lorenz gauge, $\partial_\mu A^\mu = 0$, with additional gauge freedom $\partial_\mu \partial^\mu \alpha = 0$.

Quantisation of A in the Coulomb gauge.

$$\pi^0 = \frac{\partial \mathcal{L}}{\partial \dot{A}_0} = 0, \quad \pi^i = \frac{\partial \mathcal{L}}{\partial \dot{A}_i} = -F^{0i} = E^i,$$

$$H = \int d^3x [\pi^i \dot{A}_i - \mathcal{L}] = \int d^3x \left[\frac{1}{2}(\vec{E}^2 + \vec{B}^2) - A_0(\nabla \cdot \vec{E}) \right],$$

where A_0 acts as a Lagrange multiplier. With $\square \vec{A} = 0$ and $\nabla \cdot \vec{A} = 0$ (transverse), $\vec{A}(x) = \vec{\varepsilon}(p)e^{-ip \cdot x}$ with $\vec{p} \cdot \vec{\varepsilon} = 0$: $\vec{\varepsilon}$ is the **polarisation vector**. For $\vec{p} = (0, 0, p)$, circular polarisations are $\vec{\varepsilon}_{1,2} = \frac{1}{\sqrt{2}}(1, \pm i, 0)$. Then

$$\begin{aligned} \vec{A}(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{\lambda} \left(\vec{\varepsilon}_{\lambda}(\vec{p}) a_{\vec{p}}^{\lambda} e^{i\vec{p} \cdot \vec{x}} + \vec{\varepsilon}_{\lambda}^*(\vec{p}) a_{\vec{p}}^{\lambda\dagger} e^{-i\vec{p} \cdot \vec{x}} \right), \\ \vec{E}(\vec{x}) &= \int \frac{d^3p}{(2\pi)^3} (-i) \sqrt{\frac{E_{\vec{p}}}{2}} \sum_{\lambda} \left(\vec{\varepsilon}_{\lambda}(\vec{p}) a_{\vec{p}}^{\lambda} e^{i\vec{p} \cdot \vec{x}} - \text{h.c.} \right), \end{aligned}$$

with $[a_{\vec{p}}^{\lambda}, a_{\vec{p}'}^{\lambda'\dagger}] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{p}') \delta^{\lambda\lambda'}$, so that

$$[A^i(t, \vec{x}), E^j(t, \vec{y})] = i \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p} \cdot (\vec{x} - \vec{y})} \left(\delta^{ij} - \frac{p^i p^j}{|\vec{p}|^2} \right) =: i \delta_{\perp}^{ij}(\vec{x} - \vec{y}), \quad \sum_{\lambda} \varepsilon_{\lambda}^i \varepsilon_{\lambda}^{j*} = \delta^{ij} - \frac{p^i p^j}{|\vec{p}|^2}.$$

Then

$$:H: = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda} E_{\vec{p}} a_{\vec{p}}^{\lambda\dagger} a_{\vec{p}}^{\lambda}, \quad :\vec{P}: = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda} \vec{p} a_{\vec{p}}^{\lambda\dagger} a_{\vec{p}}^{\lambda}, \quad E_{\vec{p}} = |\vec{p}|,$$

and the propagator in the Coulomb gauge is

$$\Delta_{\perp}^{ij}(x - y) := \langle 0 | T \{ A^i(x) A^j(y) \} | 0 \rangle = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 + i\epsilon} \left(\delta^{ij} - \frac{p^i p^j}{|\vec{p}|^2} \right) e^{-ip \cdot (x - y)},$$

while in the Lorenz (Feynman) gauge $\Delta_F^{\mu\nu}(p) = \frac{-i\eta^{\mu\nu}}{p^2 + i\epsilon}$.

Coupling to a current. With $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_{\mu}J^{\mu} + \mathcal{L}_m$, in the Coulomb gauge

$$\mathcal{L} = \frac{1}{2}\dot{A}_i^2 - \frac{1}{2}(\partial_i A_j)^2 - \frac{1}{2}A_0 \nabla^2 A_0 - A_0 J^0 + A_i J^i.$$

Now $\nabla^2 A_0 = -J^0$, so A_0 cannot be set to zero this time:

$$A_0(\vec{x}) = \int d^3y \frac{J^0(\vec{y})}{4\pi|\vec{x} - \vec{y}|}, \quad \mathcal{L} = \frac{1}{2}(\partial_i A_i)^2 + A_i J^i + \frac{1}{2} \int d^3y \frac{J^0(\vec{x})J^0(\vec{y})}{4\pi|\vec{x} - \vec{y}|},$$

where the last is a non-local term. Correspondingly

$$\Delta_F^{00}(x - y) = \frac{\delta(x^0 - y^0)}{4\pi|\vec{x} - \vec{y}|} = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip \cdot (x - y)}}{|\vec{p}|^2}, \quad \Delta_F^{\mu\nu}(p) = \begin{cases} i/|\vec{p}|^2, & \mu = \nu = 0, \\ \Delta_{\perp}^{ij}(p), & \mu, \nu = i, j, \\ 0, & \text{otherwise,} \end{cases}$$

i.e. $\Delta_F^{\mu\nu}(p) = \frac{i}{p^2 + i\epsilon} \left(P_{\perp}^{\mu\nu} + \frac{p^2}{|\vec{p}|^2} n^{\mu} n^{\nu} \right)$ with $n^{\mu} = (1, 0, 0, 0)$, $P_{\perp}^{00} = P_{\perp}^{0i} = P_{\perp}^{i0} = 0$, $P_{\perp}^{ij} = \delta^{ij} - p^i p^j / |\vec{p}|^2$. Contracted with conserved currents the extra terms drop out:

$$J_{\mu}^*(p) \Delta_F^{\mu\nu}(p) J_{\nu}(p) = \frac{-i}{p^2} J_{\mu}^*(p) J^{\mu}(p) \implies \Delta_F^{\mu\nu}(p) = \frac{-i\eta^{\mu\nu}}{p^2 + i\epsilon}.$$

9.7 Discrete symmetries for fermions

C (charge conjugation), P (parity), T (time reversal).

Statement (CPT theorem). All relativistic quantum field theories are invariant under the CPT transformation.

- Charge conjugation (particles $\rightarrow$ antiparticles): $Cb_p^s C = c_p^s$, $Cc_p^s C = b_p^s$, so $\psi \rightarrow -i\gamma^2\psi^*$.
- Parity ($\vec{x} \rightarrow -\vec{x}$): $Pb_{\vec{p}}^s P = b_{-\vec{p}}^s$, $Pc_{\vec{p}}^s P = -c_{-\vec{p}}^s$, so $\psi(t, \vec{x}) \rightarrow \gamma^0\psi(t, -\vec{x})$ (exchanging $\psi_L \leftrightarrow \psi_R$).
- Time reversal ($t \rightarrow -t$, $\partial_t \rightarrow -\partial_t$): T must be antilinear, $Tz = z^*T$, and it reverses the momentum (under complex conjugation $\gamma^{0,1,3} \rightarrow \gamma^{0,1,3}$, $\gamma^2 \rightarrow -\gamma^2$): $Tb_{\vec{p}}^s T = b_{-\vec{p}}^{-s}$, $Tc_{\vec{p}}^s T = c_{-\vec{p}}^{-s}$, so $\psi(t, \vec{x}) \rightarrow \gamma^1\gamma^3\psi(-t, \vec{x})$.
- CPT: $\psi(x) \rightarrow -\gamma^5\psi^*(-x)$.

10 Quantum Electrodynamics

(“The strange theory of light and matter”.)

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_\mu J^\mu, \quad \partial_\mu F^{\mu\nu} = J^\nu \implies \partial_\nu J^\nu = 0 :$$

current conservation. Under $A_\mu \rightarrow A_\mu + \partial_\mu\alpha$, $\delta S = \int d^4x \alpha \partial_\mu J^\mu$: the action is gauge invariant if $\partial_\mu J^\mu = 0$ (if J^μ is conserved).

Gauging the phase symmetry of fermions. Phase rotations $\psi \rightarrow e^{i\beta}\psi$ are a symmetry. What if this transformation were local, $\psi \rightarrow e^{i\beta(x)}\psi$? $\bar{\psi}\psi$ is still invariant, but

$$\partial_\mu\psi \rightarrow \partial_\mu(e^{i\beta(x)}\psi) = e^{i\beta(x)}[\partial_\mu\psi + i(\partial_\mu\beta)\psi], \quad \delta S_{\text{Dirac}} = - \int d^4x (\partial_\mu\beta)\bar{\psi}\gamma^\mu\psi.$$

Let us redefine $\partial_\mu \rightarrow D_\mu = \partial_\mu + iqA_\mu$ (the **minimal coupling** procedure). Under

$$A_\mu \rightarrow A_\mu + \partial_\mu\alpha, \quad \psi \rightarrow e^{i\beta(x)}\psi,$$

the action is invariant if $\beta = -q\alpha$. Then

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi = \bar{\psi}(i\not{\partial} - m)\psi - qA_\mu \underbrace{\bar{\psi}\gamma^\mu\psi}_{j^\mu}, \quad :Q: = q \int \frac{d^3p}{(2\pi)^3} \sum_s [b_{\vec{p}}^{s\dagger} b_{\vec{p}}^s - c_{\vec{p}}^{s\dagger} c_{\vec{p}}^s],$$

where q is the electric charge. The **fine-structure constant** $\alpha = e^2/4\pi \simeq 1/137$ is a small number.

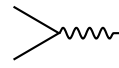
Scalar QED. Similarly, for a complex scalar

$$\begin{aligned} \partial_\mu\varphi^*\partial^\mu\varphi - m^2\varphi^*\varphi &\longrightarrow D_\mu\varphi(D^\mu\varphi)^* - m^2\varphi^*\varphi \\ &= \partial_\mu\varphi\partial^\mu\varphi^* - m^2\varphi^*\varphi + ieA_\mu \underbrace{(\varphi\partial^\mu\varphi^* - \varphi^*\partial^\mu\varphi)}_{j^\mu} + e^2\varphi\varphi^*A_\mu A^\mu. \end{aligned}$$

QED Lagrangian.

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{D} - m)\psi = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\mathcal{L}_0} + \underbrace{\bar{\psi}(i\not{\partial} - m)\psi - qA_\mu\bar{\psi}\gamma^\mu\psi}_{\mathcal{L}_{\text{int}}}.$$

Feynman rules (momentum space):

	$\frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}$	fermion propagator
	$\frac{-i\eta_{\mu\nu}}{p^2 + i\epsilon}$	photon propagator
	$-ie\gamma^\mu$	QED vertex
external lines	$u^s(p), \bar{u}^s(p)$	fermion in, out
	$\bar{v}^s(p), v^s(p)$	antifermion in, out
	$\varepsilon_\mu^\lambda(p), \varepsilon_\mu^{\lambda*}(p)$	photon in, out

10.1 Some processes

$e^-e^- \rightarrow e^-e^-$.

$$\begin{aligned}
 i\mathcal{A}_t &= \bar{u}^{s_3}(p_3)(-ie\gamma^\mu)u^{s_1}(p_1) \frac{-i\eta_{\mu\nu}}{(p_1 - p_3)^2} \bar{u}^{s_4}(p_4)(-ie\gamma^\nu)u^{s_2}(p_2) \\
 &= \frac{ie^2 [\bar{u}^{s_3}(p_3)\gamma^\mu u^{s_1}(p_1)] [\bar{u}^{s_4}(p_4)\gamma_\mu u^{s_2}(p_2)]}{(p_1 - p_3)^2}, \\
 i\mathcal{A}_u &= \frac{-ie^2 [\bar{u}^{s_4}(p_4)\gamma^\mu u^{s_1}(p_1)] [\bar{u}^{s_3}(p_3)\gamma_\mu u^{s_2}(p_2)]}{(p_1 - p_4)^2}.
 \end{aligned}$$

$e^-e^+ \rightarrow \gamma\gamma$.

$$i\mathcal{A} = -ie^2 \bar{v}^{s_2}(p_2)\gamma^\mu \varepsilon_\mu^*(p_4) \frac{\not{p}_1 - \not{p}_3 + m}{(p_1 - p_3)^2 - m^2} \gamma^\nu u^{s_1}(p_1) \varepsilon_\nu^*(p_3) + (3 \leftrightarrow 4).$$

$e^-e^+ \rightarrow e^-e^+$. t -channel ($q = p_1 - p_3$) and s -channel ($q = p_1 + p_2$):

$$\begin{aligned}
 t: & \quad \bar{u}^{s_3}(p_3)(-ie\gamma^\mu)u^{s_1}(p_1) \frac{-i\eta_{\mu\nu}}{(p_1 - p_3)^2} \bar{v}^{s_2}(p_2)(-ie\gamma^\nu)v^{s_4}(p_4), \\
 s: & \quad \frac{ie^2 [\bar{u}^{s_3}(p_3)\gamma^\mu v^{s_4}(p_4)] [\bar{v}^{s_2}(p_2)\gamma_\mu u^{s_1}(p_1)]}{(p_1 + p_2)^2}.
 \end{aligned}$$

$e^-\mu^- \rightarrow e^-\mu^-$. Only the t -channel, $q = p_1 - p_3$:

$$i\mathcal{A}_t = \frac{ie^2}{q^2} [\bar{u}_e(p_3)\gamma^\mu u_e(p_1)] [\bar{u}_\mu(p_4)\gamma_\mu u_\mu(p_2)], \quad |\mathcal{A}|^2 = \frac{e^4}{q^4} L_{(e)}^{\mu\nu} L_{(\mu)\mu\nu},$$

with $L_{(e)}^{\mu\nu} = [\bar{u}_e(p_3)\gamma^\mu u_e(p_1)][\bar{u}_e(p_3)\gamma^\nu u_e(p_1)]^*$ and similarly for the muon. Averaging over initial and summing over final spins, with $\sum_s u^s(p)\bar{u}^s(p) = \not{p} + m$,

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{A}|^2 = \frac{e^4}{4q^4} \text{Tr}[\gamma^\mu(\not{p}_1 + m_e)\gamma^\nu(\not{p}_3 + m_e)] \text{Tr}[\gamma_\mu(\not{p}_2 + m_\mu)\gamma_\nu(\not{p}_4 + m_\mu)].$$

Using the trace identities

$$\text{Tr}\{\not{a}\not{b}\} = 4a \cdot b, \quad \text{Tr}\{\not{a}\not{b}\not{c}\not{d}\} = 4(a \cdot b c \cdot d - a \cdot c b \cdot d + a \cdot d b \cdot c), \quad \text{Tr}\{\text{odd number of } \gamma\text{'s}\} = 0,$$

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{A}|^2 = \frac{2e^4}{t^2} \left[s^2 + u^2 - 4(m_e^2 + m_\mu^2)(s + u) + 6(m_e^2 + m_\mu^2)^2 \right].$$

In the high-energy limit $s, u \gg m_e^2, m_\mu^2$, with $d\sigma/d\Omega = |\mathcal{A}|^2/64\pi^2 s$,

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{32\pi^2 s} \frac{s^2 + u^2}{t^2}.$$